

SIMATS ENGINEERING

ASSESSMENT TOOL 1

**COURSE CODE : ITA 0601
LEARNING**

COURSE NAME: MACHINE

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

*I N. Harshavardhan (192324189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

N. Harshavardhan

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

1, given

- * Weight for study Hours $w_1 = 0.6$
- * Weight for Attendance $w_2 = 0.4$
- Bias, $b = -0.5$
- * Study Hours $x_1 = 2$
- * Attendance $x_2 = 1$

Step 1:- Calculate Net input

$$\begin{aligned}\text{Net input} &= (w_1 x_1) + (w_2 x_2) + b \\ &= (0.6 \times 2) + (0.4 \times 1) - 0.5 \\ &= 1.2 + 0.4 - 0.5 = 1.1\end{aligned}$$

Step 2:- Apply step Activation function

step function:

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Since $1.1 \geq 0$

Conclusion:-

- * Net input $= 1.1$
- * final output $= 1$
- * Student Passes

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

2. Back Propagation weight Update

Given:

* input weight (w) = 0.5

* learning rate (η) = 0.1

* input (x) = 2

* Actual output = 0.6

* Target output = 1

Step 1: Calculate Error

$$\text{Error} = \text{Target} - \text{Actual}$$

$$e = 1 - 0.6 = 0.4$$

Step 2: Calculate weight update

$$\Delta w = \eta \times \text{Error} \times x$$

$$= 0.1 \times 0.4 \times 2$$

$$= 0.08$$

Step 3: Update weight

$$w_{\text{new}} = w + \Delta w$$

$$= 0.5 + 0.08 = 0.58$$

final Answer

* Error = 0.4

* weight update = 0.08

* update weight = 0.58

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

3. Confusion matrix Performance

Given

- * TP = 180
- * TN = 150
- * FP = 20
- * FN = 50

a, Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{180 + 150}{180 + 150 + 20 + 50} = \frac{330}{400} = 0.825$$

Accuracy = 82.5%

b, Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{180}{180 + 20} = \frac{180}{200} = 0.90$$

Precision = 90%

c, Sensitivity

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{180}{180 + 50} = \frac{180}{230} = 0.7826$$

Recall = 78.26%

d, Specificity

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{150}{150 + 20} = \frac{150}{170} = 0.8824$$

Specificity = 88.24%

Conclusion

Accuracy = 82.5%

Precision = 90%

Recall = 78.26%

Specificity = 88.24%

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

4. Genetic Algorithm

Given

individual	Fitness $f(x) = x^2$
2	4
4	16
6	36
8	64

Step 1: Best Candidates

Highest fitness values:

- * 8 $\rightarrow$ 64
- * 6 $\rightarrow$ 36

These are selected for crossover

Step 2: Binary Representation

6 = 0110

8 = 1000

Exchange test significant Bit (LSB)

Before:

- * 0110
- * 1000

Since both LSBs are 0, the offspring remain:

- * 0110 (6)
- * 1000 (8)

off spring

* off spring 1 = 6

* off spring 2 = 8

Mutation importance

* maintains Population diversity

* Prevents premature convergence

* Helps escape local optimum solutions

* improves the chance of finding the global optimum

5. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

5 Training and Validation Accuracy

observation

* Training Accuracy = 60% $\rightarrow$ 95%

* Validation Accuracy = 58% $\rightarrow$ 72% $\rightarrow$ 68%

Issue identified

The model is suffering from overfitting

Why overfitting occurs

The neural network learns the training data very well, including noise and unnecessary patterns.

Techniques to Reduce overfitting

1. Dropout

* Randomly deactivates neurons during training

* Prevents the network from relying from heaving on specific neurons

2. Early stopping

* stops - training when validation accuracy begins to decrease

* Prevents the model from memorizing the training data.

Significance

* Measures the model performance on unseen data

* indicates the model ability to generalize

* Helps detect overfitting and underfitting

